MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 7363

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, xenon gas

See figure 1

Mounting position: Any

Weight: 3 ounces (85 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	epy	[illegible]	ib	tk	TA
Unit:	Vac	v	v	mdc	a	sec	°C
Maximum:	2.63	1,500	1,500	1.6	20	---	+75
Minimum:	2.37	---	---	---	---	10	-55
TEST CONDITIONS:	2.5	1,500	1,500	---	---	10	---

GENERAL:

Qualification - Required

Ⓔ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3201	Critical grid voltage for conduction	-	Ebb = 1,500 Vdc; Rg = 10,000 ohms; Rp = 950 to 5,000 ohms	0.65	II	Ecc	-5.0	-10.0	Vdc
(E)3201	Critical anode voltage for conduction	-	Ecc = +3.0 Vdc; Rg = 1,000 ohms; Rp = 300 to 5,000 ohms	0.65	II	Ebb	---	50	Vdc
(E)3204	Voltage drop	1	Ib = 1.6 Adc; Epp = 130 Vac	0.65	II	Etd	---	14.0	Vdc
3206	Operation	-	epx = epy = 1,500 v; Ib = 1.6 Adc; Rg = 10,000 ohms	0.65	II	Ecc	-5.5	-10.5	Vdc
3216	Grid current (1)	2,8	epx = epy = 1,500 v; Ib = 1.6 Adc; Rg = 10,000 ohms and 1 megohm; t = 150 seconds	0.65	II	Ic	---	2.0	µAdc
1231	Thyratron emission	3,8	ib = 20 a	0.65	II	etd	---	30.0	v
(F)3214	High-frequency operation	5	Epp = 125 Vac; R1/Ib = 1.6 Adc	0.65	II	F	3,400	---	Hz
	Quality conformance inspection, part 2								
1031	Variable-frequency vibration	-	No voltages applied	---	---	---	---	---	---
3241	Heater current	-		---	---	If	7.5	9.5	Aac
---	No-load operation	4	Ep = 0	---	---	t	100	---	hrs
3216	Grid current (2)	2	epx = epy = 1,500 v; Ib = 1.6 Adc; Rg = 10,000 ohms and 1 megohm; t = 300 seconds	---	---	Ic	---	2.0	µAdc
	Quality conformance inspection, part 3								
---	Life test	6,7	Group C; ib = 20 a; Rg = 10,000 ohms; Ecc = -45 Vdc; egy = 60 v; t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
1231	Thyratron emission	-		---	---	etd	---	35	v
3206	Operation	-		---	---	Ecc	-5.0	-11.0	Vdc
3216	Grid current (1)	-		---	---	Ic	---	3.0	µAdc
1031	Vibration fatigue	9	No voltages applied	---	---	---	---	---	---
1041	Shock	9	360 G; no voltages applied	---	---	---	---	---	---
---	Shock-test end points:								
3206	Operation	-		---	---	Ecc	-5.5	-10.5	Vdc
3241	Heater current	-		---	---	If	7.5	9.5	Aac

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NOTES:

1. Wattmeter method. With an alternating current source and resistive load, measure the power loss in the tube with a wattmeter so that the current coil is in series with, and the voltage coil in parallel with, the tube. The tube voltage drop (Etd) is the total power loss minus the coil loss under load, divided by the average or dc current (Ib).

2. Not more than 15 seconds of operation shall separate measurements at 10,000 ohms and 1 megohm grid resistance.

3. Connect grid to anode through 10,000 ohms. Adjust Epp and Rp to give required ib.

4. After no-load operation, tubes shall be within limits specified herein for operation and grid-current tests. In operation test, the operating time shall be a minimum of 30 and a maximum of 120 seconds.

(E)5. Test shall be taken between a minimum operating time of 15 seconds and a maximum of 60 seconds. The circuit parameters are the following: Cg = 250 pF; Rg = 47,000 ohms; Ecc1 = 22.5 ±2 Vdc; Ecc2 = 45 ±2 Vdc.

6. Tubes shall operate in a single phase, resistive load, full-wave rectifier for 25 ±3 percent of every 15 second recurring period. The resistive load should be adjusted for an average current of 6.4 Adc per tube during the "on" period. (See figure 2 for typical circuit).

7. The grid pulse shall be obtained from a peaking transformer with the source impedance not to exceed 10,000 ohms.

8. This test to be performed at the conclusion of the holding period.

(E)9. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

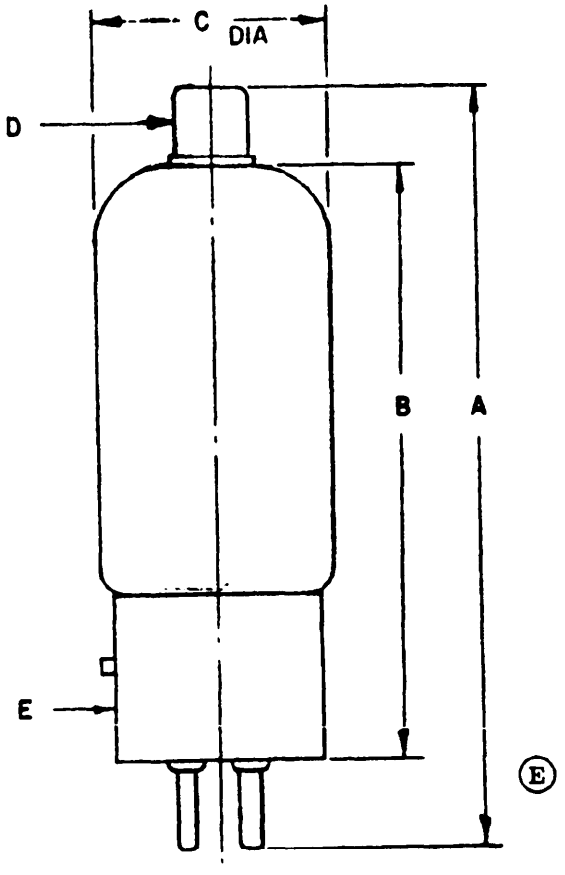
User activities:
Army - MU
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2990)

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PIN CONNECTIONS	
Pin No.	Element
1	f
2	f (c.t.)
3	g
4	f
cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	4.625 (117.48)	5.000 (127.00)
B	3.375 (85.73)	3.625 (92.08)
C		1.563 (39.70)
Quality conformance inspection, part 3 (see note a)		
D	Cap: C1-5 (EIA)	
E	Base: A4-10 (EIA) (see note b)	

NOTES:

a. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

b. Metal shell, isolantite insert.

FIGURE 1. Outline drawing of electron tube type 7363.

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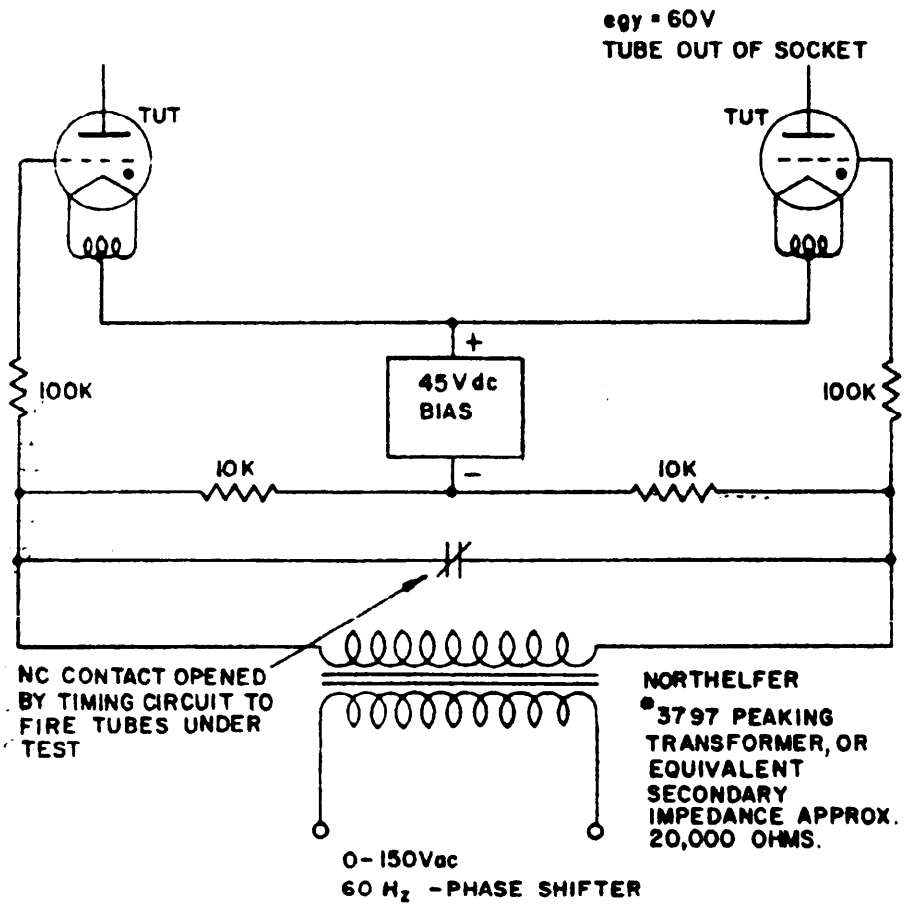


FIGURE 2. Typical circuit for life testing of electron tube type 7363 with resistive load.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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NAVY DEPARTMENT
DOD 316

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Commente and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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